

一. 1. $\frac{7}{15}$ 2. $f_Y(y) = \begin{cases} \frac{1}{2y}, & e^{-1} < y < e \\ 0, & \text{其他} \end{cases}$ 3. 0.5 4. (19.5617, 20.4383) 5. 5

(3分/题, 总共 15分)

二. 1.D 2.B 3.A 4.C 5.B

(3分/题, 总共 15分)

三. 解: 设 A = “先取出的为一等品”,

B = “后取出的为一等品”, $C_i =$ “取出的为第 i 箱”, $i=1,2$.

(1) $P(A) = P(A|C_1)P(C_1) + P(A|C_2)P(C_2) = \frac{1}{2}(\frac{10}{50} + \frac{18}{30}) = \frac{2}{5}$, 5分

(2)

$$\begin{aligned} P(B|A) &= \frac{P(AB)}{P(A)} = \frac{P(AB(C_1+C_2))}{P(A)} = \frac{P(C_1AB) + P(C_2AB)}{P(A)} \\ &= \frac{P(C_1)P(A|C_1)P(B|AC_1) + P(C_2)P(A|C_2)P(B|AC_2)}{P(A)} \\ &= \frac{1}{2}(\frac{10}{50} \times \frac{9}{49} + \frac{18}{30} \times \frac{17}{29}) / \frac{2}{5} = \frac{690}{1421} = 0.48557, \end{aligned}$$

4分

四. 解(1)总体 X 的概率密度为 $f(x) = \begin{cases} \frac{1}{\theta-1}, & 1 \leq x \leq \theta \\ 0, & \text{其他} \end{cases}$,

分布函数为 $F(x) = \begin{cases} 0, & x < 1 \\ \frac{x-1}{\theta-1}, & 1 \leq x \leq \theta \\ 1, & x > \theta \end{cases}$

$X_{(n)}$ 的分布函数为: $F_{X_{(n)}}(x) = [F(x)]^n$

故 $f_{X_{(n)}}(x) = n[F(x)]^{n-1}f(x) = \begin{cases} \frac{n(x-1)^{n-1}}{(\theta-1)^n}, & 1 \leq x \leq \theta \\ 0, & \text{其他} \end{cases}$ 5分

$$\begin{aligned} (2) EX_{(n)} &= \int_1^\theta x \frac{n(x-1)^{n-1}}{(\theta-1)^n} dx = \frac{n}{(\theta-1)^n} [\int_1^\theta (x-1)^n dx + \int_1^\theta (x-1)^{n-1} dx] \\ &= \frac{n}{(\theta-1)^n} [\frac{(\theta-1)^{n+1}}{n+1} + \frac{(\theta-1)^n}{n}] = \frac{n(\theta-1)}{n+1} + 1 = \frac{n}{n+1}\theta + \frac{1}{n+1} \end{aligned}$$

$$EX_{(n)}^2 = \int_1^\theta x^2 \frac{n(x-1)^{n-1}}{(\theta-1)^n} dx$$

$$= \frac{n}{(\theta-1)^n} \left[\int_1^\theta (x-1)^{n+1} dx + \int_1^\theta 2x(x-1)^{n-1} dx - \int_1^\theta (x-1)^{n-1} dx \right]$$

$$= \frac{n}{(\theta-1)^n} \left[\frac{(\theta-1)^{n+2}}{n+2} \right] + \frac{2n(\theta-1)}{n+1} + 2 - 1 = \frac{n(\theta-1)^2}{n+2} + \frac{2n(\theta-1)}{n+1} + 1$$

$$DX_{(n)}^2 = EX_{(n)}^2 - (EX_{(n)})^2 = \frac{n(\theta-1)^2}{n+2} + \frac{2n(\theta-1)}{n+1} + 1 - \left(\frac{n(\theta-1)}{n+1} + 1 \right)^2$$

$$= \frac{n(\theta-1)^2}{n+2} - \frac{n^2(\theta-1)^2}{(n+1)^2} = \frac{n}{(n+2)(n+1)^2} (\theta-1)^2 \quad 4 \text{ 分}$$

五. 解: (1) 因 $\max(X, Y) = Y$

$$\text{故 } f_M(y) = f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_0^y x e^{-y} dx, & y > 0 \\ 0, & y \leq 0 \end{cases} = \begin{cases} \frac{1}{2} y^2 e^{-y} & y > 0 \\ 0, & y \leq 0 \end{cases},$$

令解: 令 M 的分布函数 $F_M(z)$,

$$F_M(z) = P(\max(X, Y) \leq z)$$

当 $z \leq 0$ 时, $F_M(z) = 0$,

$$\text{当 } z > 0 \text{ 时, } F_M(z) = P(X \leq z, Y \leq z) = \int_0^z \left(\int_x^z x e^{-y} dy \right) dx = 1 - e^{-z} - z e^{-z} - z^2 e^{-z} / 2.$$

$$\text{故 } f_M(z) = F'_M(z) = \begin{cases} \frac{1}{2} z^2 e^{-z} & z > 0 \\ 0, & z \leq 0 \end{cases} \quad 3 \text{ 分}$$

(2) 因 $Z = \max(X, Y) + \min(X, Y) = X + Y$

$$\text{故 } f_z(z) = \int_{-\infty}^{+\infty} f(x, z-x) dx$$

$$\text{使 } f(x, z-x) \text{ 不为 } 0 \text{ 的区域为: } 0 < x < z-x \Leftrightarrow \begin{cases} x > 0 \\ z > 2x \end{cases}.$$

当 $z \leq 0$ 时, $f_z(z) = 0$,

$$\text{当 } z > 0 \text{ 时, } f_z(z) = \int_0^{\frac{z}{2}} x e^{-(z-x)} dx = e^{-z} + \frac{z}{2} e^{-z/2} - e^{-z/2}$$

$$\text{故 } f_z(z) = \begin{cases} e^{-z} + \left(\frac{z}{2} - 1\right) e^{-z/2} & z > 0 \\ 0, & z \leq 0 \end{cases} \quad 3 \text{ 分}$$

令解: $f_z(z) = \int_{-\infty}^{+\infty} f(z-y, y)dy$, 不为零的区域 $0 < z-y < y \Rightarrow \begin{cases} z > y \\ z < 2y \end{cases}$;

$$(3) P(X+Y \leq 1) = \int_0^{\frac{1}{2}} dx \int_x^{1-x} xe^{-y} dy = 1 - e^{-\frac{1}{2}} - e^{-1}. \quad 3 \text{ 分}$$

六. 解: (1) 矩估计: 两次分部积分可得

$$\begin{aligned} EX &= \int_0^{\infty} \lambda^2 x^2 e^{-\lambda x} dx \\ &= \int_0^{\infty} -\lambda x^2 de^{-\lambda x} = -\lambda x^2 e^{-\lambda x} \Big|_0^{\infty} + 2\lambda \int_0^{\infty} xe^{-\lambda x} dx = 2 \int_0^{\infty} -x de^{-\lambda x} = 2(-xe^{-\lambda x} \Big|_0^{\infty} + (-\frac{1}{\lambda} e^{-\lambda x} \Big|_0^{\infty})) = \frac{2}{\lambda} \end{aligned}$$

解得 $\lambda = \frac{2}{EX}$

故参数 λ 的矩估计量为: $\hat{\lambda}_1 = \frac{2}{\bar{X}}$; 4 分

(2) 最大似然估计:

$$\begin{aligned} \text{似然函数为 } L(x_1, x_2, \dots, x_n; \lambda) &= \prod_{i=1}^n f(x_i, \lambda) = \prod_{i=1}^n \lambda^2 x_i e^{-\lambda x_i}, x_i > 0 \\ &= \lambda^{2n} e^{-\lambda \sum_{i=1}^n x_i} \prod_{i=1}^n x_i \end{aligned}$$

对数似然:

$$\ln L(x_1, x_2, \dots, x_n; \lambda) = 2n \ln \lambda + \sum_{i=1}^n \ln x_i - \lambda \sum_{i=1}^n x_i$$

$$\text{令 } \frac{d \ln L}{d \lambda} = \frac{2n}{\lambda} - \sum_{i=1}^n x_i = 0,$$

故参数 λ 的最大似然估计量 $\hat{\lambda}_2 = \frac{2}{\bar{X}}$ 5 分

七. 解: (1) 设 X 的分布函数为 $F_X(x)$, 即 $F_X(x) = P(X \leq x)$, 则

当 $x \leq 0$ 时, $F_X(x) = 0$,

当 $x > 0$ 时,

$$F_X(x) = P(X \leq x) = 1 - P(X > x) = 1 - P(N(x) = 0) = 1 - \frac{(\lambda x)^0}{0!} e^{-\lambda x} = 1 - e^{-\lambda x},$$

故 X 的分布函数为: $F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x > 0 \\ 0, & x \leq 0 \end{cases}$,

X 的概率密度函数为: $f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0, & x \leq 0 \end{cases}$ 2 分

即 X 服从参数为 λ 的指数分布.

(2) $P(X > 2 | X > 1) = P(X > 1) = 1 - F_X(1) = e^{-\lambda}$ 2 分 (据指数分布具有后效性特点)